

DevOps Series

Ansible Deployment of Sensu and Uchiwa

While Sensu is a free and open source monitoring and telemetry tool, Uchiwa provides the free and open source dashboard for Sensu. This article covers the Ansible deployment of Sensu and Uchiwa.



Sensu Core is a free and open source monitoring and telemetry solution. You can use it to monitor services, an application's health, servers and important KPIs. It is primarily a monitoring event pipeline which allows you to filter or amend incoming events, and send alerts and notifications. It uses JSON for all its configuration files and integrates well with automation tools. Uchiwa is a free and open source dashboard for Sensu, written in the Go programming language. It also requires Node.js for front-end assets, and JavaScript. Both Sensu Core and Uchiwa are released under the MIT licence.

Setting them up

A CentOS 7 (x86_64) guest virtual machine (VM) using KVM/QEMU will be used to set up Sensu.

The host system is a Parabola GNU/Linux-libre x86_64 system and Ansible is installed using the distribution package manager. The version of Ansible used is 2.6.0 as indicated below:

```
$ ansible --version
ansible 2.6.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.6/site-
packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

The Ansible inventory, playbook and configuration files are created on the host system as follows:

```
ansible/inventory/kvm/
```

```
/playbooks/configuration/
/files/
```

The `inventory/kvm/inventory` file contains the following:

```
sensu ansible_host=192.168.122.43 ansible_connection=ssh
ansible_user=centos ansible_password=centos123
```

A 'centos' user is created in the guest VM and sudo access is provided for this user using the 'visudo' command. SELinux needs to allow access to port 3000 for the Uchiwa dashboard. You should also add an entry in `/etc/hosts` file for the CentOS VM as shown below:

```
192.168.122.43 sensu
```

You can now test connectivity from Ansible to the CentOS 7 VM using the following command:

```
$ ansible -i inventory/kvm/inventory sensu -m ping
sensu | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Installing the repositories

The first step is to install the repositories required for Sensu. The Extra Packages for Enterprise Linux (EPEL) repository is added. The `sensu.repo` also needs to be created in the guest VM, whose file contents are shown below:

```
[sensu]
name=sensu
baseurl=https://sensu.global.ssl.fastly.net/
yum/$releasever/$basearch/
```